

THE 3-GATE MORPHOLOGICAL REACHABILITY & SINGULARITY VALIDATOR

Deterministic screening of untrusted VLM stochastic proposals before granting physical motor execution authority

UNTRUSTED INPUT PROPOSAL (VLM /LLM SYSTEM 2)

Candidate Target Pose: $T_{target} = [R \mid p] \in SE(3)$ | Payload: $m = 0.50$ kg | Origin: Linux MPU (Untrusted)



GATE 1: WORKSPACE GEOMETRY (W_{reach})

✓ ~~BASED~~ WITHIN DEXTEROUS WORKSPACE

$$0.15 \text{ m} \leq r \leq 0.85 \text{ m}$$

$$z_{target} \geq z_{table} + 20 \text{ mm}$$

- **Workspace Sphere:** Rejects targets beyond mechanical reach.
- **Table Plane Hull:** Prevents plunging into supporting surface
- **Joint Range Check:** Validates $q_{min} \leq IK(T) \leq q_{max}$.

GATE 2: SINGULARITY & MANIPULABILITY

✓ ~~BASED~~ CONDITION NUMBER $\kappa(J) = 14.2$

$$w(q) = \sqrt{(\det(J \cdot J^T))} \geq 0.05$$

$$\det(J) \neq 0 \text{ (No Rank Loss)}$$

- **Singularity Avoidance:** Near singularities, small task velocities result in large joint velocities.
- **Yoshikawa Ellipsoid:** Ensures non-zero dexterity.

GATE 3: STATIC TORQUE & PAYLOAD

✓ ~~BASED~~ PEAK TORQUE 62% OF RATED LIMIT

$$|\tau_i| \leq 0.80 \cdot \tau_{continuous}$$

Thermal Margin: 20% Headroom

- **Gravity Holding:** Computes required motor torque.
- **Thermal Protection:** Prevents motor winding $I^2 R$ Joule overheating.

GATEWAY AUDIT COMPLETE: ALL 3 MORPHOLOGICAL GATES SATISFIED

- **Permission Granted:** Proposal certified as kinematically admissible, singularity-safe, and within motor thermal limits.
- **Handoff to System 1.5:** Target pose T_{target} is passed to the real-time trajectory planner to synthesize C^2 -continuous action chunks.
- **Defect Reporting:** If any gate had failed, an explicit error code (e.g. ERR_SINGULARITY_KAPPA) would be returned to System 2 for re-prompting.